TARUN REDDY
AI/ML Engineer
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PROFESSIONAL SUMMARY

- Over 4+ years of experience in AI/ML Engineering, Machine Learning, Generative AI, LLMs, Deep Learning, and AI Model Development across Healthcare, Financial Services, & Life Sciences domains.
- Proficient in Python, SQL, Java, C++, Bash, and TypeScript for developing machine learning models, data pipelines, APIs, and AI applications.
- Skilled in Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Keras, CNN, RNN, LSTM, GRU, Transformers, and NLP for predictive and deep learning solutions.
- Experienced in Generative AI, LLMs, OpenAI, Claude, Gemini, Llama, Mistral, Hugging Face, Prompt Engineering, Context Engineering, and LLM APIs.
- Strong in RAG, Semantic Search, Embeddings, Vector Databases, Knowledge Bases, Chunking, Reranking, and Hybrid Search using Pinecone, Azure AI Search, Elasticsearch, OpenSearch, Qdrant, Weaviate, and FAISS.
- Hands-on experience with LLM Fine-Tuning, LoRA, QLoRA, PEFT, RLHF, Quantization, LangChain, LangGraph, LlamaIndex, MCP, and Agentic AI.
- Proficient in AWS SageMaker, Amazon Bedrock, Azure Machine Learning, Microsoft Foundry, Google Vertex AI, Docker, Kubernetes, Terraform, MLflow, Kubeflow, and CI/CD for AI deployment and MLOps.
- Adept at developing scalable AI solutions using FastAPI, Flask, Spark, PySpark, Airflow, Kafka, Databricks, Snowflake, PostgreSQL, Redis, vLLM, and KServe, with strong problem-solving, communication, and teamwork skills.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, C++, Bash, TypeScript
- **Machine Learning**: Supervised Learning, Unsupervised Learning, Semi-Supervised Learning, Regression, Classification, Clustering, Ensemble Learning, Feature Engineering, Model Selection
- **Deep Learning:** PyTorch, TensorFlow, Keras, JAX, CNN, RNN, LSTM, GRU, Transformers, Attention Mechanisms, Transfer Learning
- **Generative AI & LLMs**: Generative AI, Large Language Models, Foundation Models, OpenAI, Anthropic Claude, Gemini, Llama, Mistral, Hugging Face, LLM APIs, Prompt Engineering
- **RAG & AI Applications:** Retrieval-Augmented Generation, Semantic Search, Hybrid Search, Embeddings, Chunking, Reranking, Context Engineering, Knowledge Bases, Query Expansion, Grounding
- **LLM Fine-Tuning & AI Agents**: LoRA, QLoRA, PEFT, RLHF, Instruction Tuning, Quantization, Model Distillation, LangChain, LangGraph, LlamaIndex, Agentic AI, Multi-Agent Systems, MCP
- **Vector Databases & ML Frameworks**: Pinecone, Azure AI Search, Elasticsearch, OpenSearch, Weaviate, Qdrant, ChromaDB, pgvector, FAISS, Scikit-learn, XGBoost, LightGBM
- **MLOps & Cloud:** MLflow, Kubeflow, Weights & Biases, AWS SageMaker, Amazon Bedrock, Azure Machine Learning, Microsoft Foundry, Google Vertex AI, Model Deployment, Model Monitoring
- **Data Engineering & Databases:** Pandas, NumPy, SciPy, Apache Spark, PySpark, Airflow, Kafka, Databricks, Snowflake, PostgreSQL, Redis, ETL, Data Pipelines
- **Deployment & DevOps:** Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins, CI/CD, FastAPI, Flask, REST APIs, Microservices, vLLM, KServe, Model Serving, GPU Computing

PROFESSIONAL WORK EXPERIENCE

Johnson & Johnson | New Brunswick, NJ | AI/ML Engineer **Oct 2025 – Present**
Responsibilities:

- Architect and implement enterprise AI/ML solutions using Python, PyTorch, Transformers, and advanced machine learning techniques to support intelligent healthcare and life-sciences applications.
- Develop production-grade Generative AI and LLM applications using foundation models, LLM APIs, prompt engineering, context engineering, and structured generation patterns for domain-specific workflows.
- Engineer Retrieval-Augmented Generation pipelines incorporating embeddings, semantic retrieval, chunking, reranking, grounding, and knowledge-base integration to improve enterprise question-answering accuracy.
- Build intelligent knowledge systems using Azure AI Search, vector indexing, hybrid retrieval, and PostgreSQL/pgvector, enabling contextual access to large volumes of organizational information.
- Develop AI agents and agentic workflows with LangGraph, LangChain, tool calling, function calling, agent memory, and MCP to automate multi-step research and decision-support processes.
- Fine-tune domain-specific language models using LoRA, QLoRA, PEFT, supervised fine-tuning, quantization, and model distillation while optimizing computational requirements for enterprise workloads.

- Implement deep learning models using Transformer architectures, attention mechanisms, transfer learning, and PyTorch for NLP and intelligent document-processing use cases.
- Processed over 5 million enterprise healthcare records and documents through AI pipelines, improving information retrieval efficiency by approximately 35% and reducing manual document discovery efforts.
- Optimized LLM and RAG workflows through retrieval tuning, prompt evaluation, and model optimization, achieving approximately 25% improvement in response relevance while reducing unnecessary inference overhead.
- Establish LLM evaluation workflows using RAGAS, DeepEval, prompt evaluation, hallucination detection, and automated quality metrics to validate model responses before production deployment.
- Apply responsible AI practices including guardrails, prompt-injection protection, data privacy, model explainability, bias detection, and AI safety controls across production AI workflows.

Environment: Python, SQL, PyTorch, Hugging Face Transformers, Generative AI, LLMs, RAG, Azure AI Search, pgvector, LangChain, LangGraph, MCP, LoRA, QLoRA, PEFT, RAGAS, DeepEval, PostgreSQL, FastAPI, Docker, Kubernetes, Azure Machine Learning, Microsoft Foundry, MLflow, GitHub Actions

JPMorgan Chase & Co. | Boston, MA | AI/ML Engineer **Dec 2024 – Oct 2025**

Responsibilities:

- Built machine learning solutions using Scikit-learn, XGBoost, and LightGBM to support financial analytics and predictive modeling.
- Performed comprehensive feature engineering, model selection, dimensionality reduction, regression, classification, clustering, and ensemble learning to improve predictive model performance.
- Built scalable data-processing workflows with Apache Spark and PySpark to transform high-volume financial datasets for machine learning and analytical applications.
- Implemented real-time and batch data pipelines using Apache Kafka and Apache Airflow, supporting reliable ingestion, transformation, validation, and downstream model processing.
- Developed cloud-native ML workflows using AWS SageMaker and Amazon Bedrock, integrating managed model development, experimentation, deployment, and foundation-model capabilities.
- Containerized machine learning services using Docker and Kubernetes, enabling repeatable deployment across development, testing, and production environments.
- Built secure REST-based inference services with FastAPI and Flask, integrating trained models into scalable microservice architectures.
- Improved model experimentation efficiency by approximately 30% through standardized MLflow experiment tracking, model versioning, and reproducible training workflows.
- Implemented data quality checks, model monitoring, data-drift and model-drift detection, supporting proactive identification of production model degradation.
- Automated deployment workflows through GitLab CI/CD, Jenkins, Terraform, and Helm, strengthening release consistency and infrastructure-as-code practices.

Environment: Python, SQL, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, SciPy, Apache Spark, PySpark, Kafka, Airflow, AWS SageMaker, Amazon Bedrock, Docker, Kubernetes, FastAPI, Flask, MLflow, GitLab CI/CD, Jenkins, Terraform, Helm, PostgreSQL, Redis

Fortis Healthcare | Gurugram, India | Software Engineer **Jan 2022 – Jul 2024**

Responsibilities:

- Developed healthcare applications and data solutions using Python, Java, SQL, REST APIs, and Microservices.
- Built machine learning models using Scikit-learn for classification, regression, clustering, feature engineering, and predictive analytics.
- Developed NLP and deep learning models using TensorFlow, Keras, CNN, RNN, LSTM, & GRU for healthcare data.
- Processed and transformed healthcare datasets using Pandas, NumPy, Apache Spark, and PySpark, including ETL/ELT and data quality validation.
- Developed backend services using Flask, PostgreSQL, Redis, and REST APIs for application integration and data access.
- Automated data and machine learning workflows using Apache Airflow and Docker.
- Collaborated with technical and business teams to troubleshoot issues, improve application performance, and deliver reliable healthcare software solutions.

Environment: Python, Java, SQL, Scikit-learn, TensorFlow, Keras, CNN, RNN, LSTM, GRU, Pandas, NumPy, Apache Spark, PySpark, Apache Airflow, PostgreSQL, Redis, Flask, REST APIs, Docker, Microservices, ETL/ELT, Data Quality

EDUCATION

- **Master degree in Computer Science** from Clark University, Massachusetts, United States.
- **Bachelor of Technology in Computer Science** from Jawaharlal Nehru Technological University Hyderabad (JNTUH), India.